

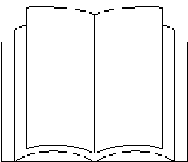


Operator's manual for Attachment



Log grab

Product number: for 200 series Avant loaders: A34822
 for 300-700 series Avant loaders: A33503



Please read this manual carefully before using the equipment, and follow all instructions.

Keep this manual for later reference.

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Appendix: EC Declaration of Conformity

This manual uses the following warning symbols to indicate factors that must be taken into account to reduce the risk of personal injury or damage to property:

	This safety symbol refers to important safety information in this manual. It warns of an immediate hazard that could cause serious personal injury or damage to property. Read the warning text accompanying the symbol carefully and ensure that other operators are also familiar with the warnings, since personal safety is at stake.
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This notice symbol indicates information about the correct operation and maintenance of the equipment.

Failure to observe the instructions accompanying the symbol can lead to equipment breakdown or other property damage.

In addition to the safety instructions included in this manual, you must observe all occupational safety regulations, local laws and other regulations concerning the use of the equipment. Due to continuous product development, some of the details in this manual may differ from your equipment. We reserve the right to change the contents of the manual without notification. This operator's manual is a translation of the original Finnish version.

1. Foreword

Avant Tecno Oy would like to thank you for your purchase of this AVANT attachment. It has been designed and manufactured on the basis of years of product development and experience. By familiarising yourself with this manual and following the instructions, you ascertain your safety and ensure the reliable operation and long service life of the equipment. Read the instructions carefully before starting to use the equipment or performing maintenance.

A separate loader model specific installation kit including hydraulic hoses and components is delivered with the attachment. The log grab must be attached to Avant pallet forks and the hydraulic kit must be assembled before the attachment can be used. Included in this manual are instructions about how to attach the log grab and how to safely install the hydraulic hoses and quick couplings or the multiconnector system. Some experience in work on hydraulic assemblies and basic tools are required. It is however recommended to leave the installation of hydraulic components to professionals.

When installing the components several important safety points must be considered. Hazards related to hydraulic installations include injection of oil into body, oil spray, or crushing caused by movements due to leaking oil. The components can be installed safely when following the instructions strictly. In doubt, contact your dealer or the nearest service point for more information.

The manual includes important instructions for experienced AVANT operators as well. Ensure that all persons using the equipment have received proper guidance and familiarised themselves with the manual and all safety instructions before using the equipment. Keep this manual at hand throughout the service life of the equipment. If the manual is lost, you can request a new one from the retailer or manufacturer. If you sell or transfer the equipment, be sure to hand over this manual to the new owner.

2. Designed purpose of use

The AVANT Log grab is an attachment intended to be used only with AVANT pallet forks. With the pallet forks that have a log grab mounted onto them you can lift and transport single logs, smaller timber bundle, or fence posts. The log grab is also suitable for lifting and moving tree branches or piles of brushwood. The grab is equipped with a powerful hydraulic cylinder and can therefore can also be used to lift bigger stones or rocks. Handling of pallets is possible even if the log grab is attached to the pallet forks, but the grapple limits the height of the pallet.



The log grab is intended to be used only when attached as instructed to Avant pallet forks A33029, A21047, or A21039.

The attachments have been designed to require as little maintenance as possible. Regular maintenance tasks can be performed by the operator. The assembling of the hydraulic components of the log grab requires some previous experience and basic tools. The assembly and the service of the hydraulic components should be left to qualified professionals, if unsure of how to perform the tasks safely. Appropriate protective equipment and original spare parts must be used for the maintenance. Read the maintenance and servicing instructions in this manual. Please contact your AVANT retailer if you have additional questions about the operation or maintenance of the equipment or if you require spare parts or maintenance services.

3. Safety instructions for the log grab

Please bear in mind that safety is the result of several factors. The loader-attachment combination is highly powerful, and improper or careless use may cause serious personal injury or property damage. Due to this, you should carefully familiarise yourself with the manuals of both the loader and the attachment before starting operation. Do not use the attachment if you have not familiarised yourself with its operation and the related hazards.

The purpose of these instructions is to help you to:

- assemble and attach the log grab safely
- use the equipment in a safe and efficient manner
- monitor and prevent any hazardous situations
- keep the equipment intact and ensure a long service life



- Follow the safety instructions laid out in the operator's manual of the pallet forks.
- Assemble the attachment according to instructions provided in this manual.
- Connect the hydraulic couplings the same way at each use to prevent unexpected operation of the log grab.
- Always transport the load as low and as close to the machine as possible to keep the centre of gravity low.
- Never leave the equipment unattended if the grapple or the pallet forks are raised.
- Do not connect any lifting equipment to the log grab.
- Familiarise yourself with the controls of the loader and the attachment at a safe area.
- Pay attention to the surroundings and any other persons and machines moving in the vicinity.
- Pay attention to the contours of the terrain and other hazards in the operating area. Note for example any possible obstacles above the equipment before lifting the load high.
- Observe instructions concerning storage to ensure that the attachment is secured against tipping over during its storage.
- Shut down the loader engine before any cleaning, maintenance or adjustments. Ensure that the equipment and the grapple are properly supported before going near the attachment.
- Keep bystanders at a safe distance from the equipment and the loader boom.
- Never use the attachment to lift persons or as any kind of access platform. The loader is not designed for lifting persons.
- Do not modify the structure of the attachment in a manner that would affect its safety.
- Also read the safety instructions for the loader in the operator's manual of the loader.

Do not try to move the pallet forks or the log grab manually. The components are heavy and may tip over, or you may injure yourself, if attempting to move them by hand. Store the equipment to a place which can be accessed directly with the loader.



Note that both the grapple of the attachment and the loader boom move down even if the engine has been stopped. Never go under the attachment, its grapple, or under a raised boom. Lower the boom and the grapple before leaving the driver's seat.



Familiarise yourself with the loader controls in a safe area. Pay particular attention to the safe stopping of the attachment and the machine. If you are unfamiliar with the controls of the loader, it is recommended to practice its use without any attachments.

Remember to wear proper protective clothing:



- The noise level in the driver's seat may exceed 85 dB(A). Wear hearing protection while working with the loader.



- The use of safety boots is recommended while working with the loader.
- Wear protective gloves when handling hydraulic components.

4. Safety instructions for installing hydraulic fittings

It is recommended to leave the installation of hydraulic components to professional service technicians. Local or company specific regulations about qualifications needed to install hydraulic components must be observed.

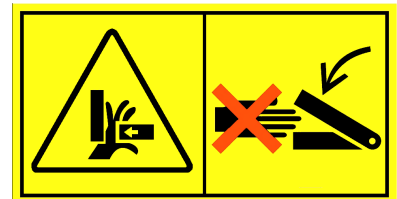
The following safety procedures must be followed when installing hydraulic components. Contact your nearest Avant dealer or service point if you have any additional questions or if you need installation services.



Before handling hydraulic components, make sure that the hydraulic system of the attachment and the loader are completely depressurized. See instructions on page 14 to release the residual pressure. High pressure fluid ejection may penetrate skin and cause serious injuries. Do not hold your hand near a fitting when tightening or opening it. Never handle pressurized fittings.



Make sure the attachment is securely lowered on the ground. Any part of the attachment must not be supported only by the hydraulic pressure or hydraulic cylinders. Do not loosen any fitting as long as any part of the attachment may move. Use additional supports as necessary. Lower the attachment securely on the ground, shut down the loader engine and lower the grapple on ground or additional supports. Depressurize the hydraulic systems completely.



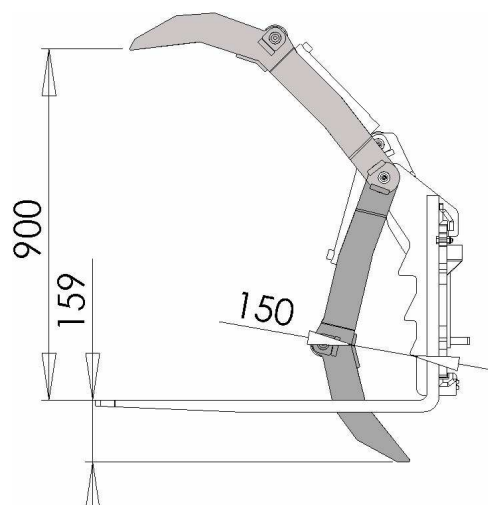
Ensure that the loader boom or any part of the attachment will not move and cause a crushing hazard:

- Lower the boom to its lowest position
- Lower the attachment firmly on the ground
- Lower any cylinder operated moving component of the attachment, i.e. the arm of the log grab. Use additional supports or lifting equipment as needed.
- Shut down the loader engine.
- Move the auxiliary hydraulics control lever to its extreme positions several times to release residual pressure.

5. Log grab specifications

Table 1 – Log grab technical data

Product number	A34822	A33503
Mounts on pallet forks, number:	A33029	A21047 A21039
Weight, complete:	60 kg	
Fork width when the log grab is mounted:	462 - 790 mm	
Assembly kit product number:	A33570*	Standard quick couplings: A44701 Multiconnector system: A45866
Maximum operating pressure:	22,5 MPa (225 bar)	
Suitable AVANT loader:	216, 218, 220	300 – 700

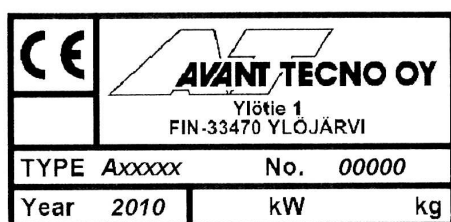


*Contact service if installing other type of hose kit to 200 series equipment.

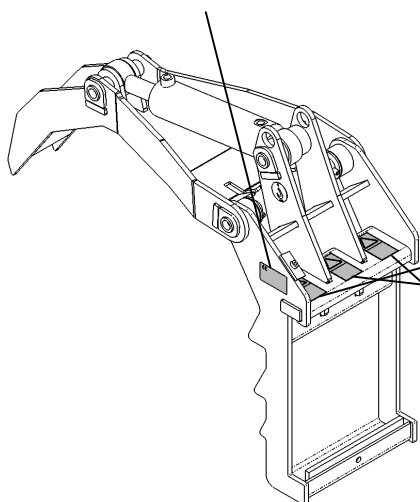
Listed below are the markings and labels fastened to the attachment, which must be visible on the equipment. Check before use, that the labels are in place and in readable condition. Replace the warning labels, if they have become unclear or detached completely. New labels are available via your retailer or contact information provided on the cover page.



The warning decals contain important safety information. Replace damaged or missing decals with new ones.



Identification plate of the attachment



A46771



A46772



A46803

Decal	Explanation
A46771	Read instructions before use.
A46772	Do not go under a raised attachment or the loader boom, keep a safe distance to lifted equipment.
A46803	Crushing hazard, keep clear of moving parts.

6. Assembling the log grab

Check that the hydraulic quick couplings in the separate installation kit are the same type as the loader quick couplings. Attach the pallet forks to the loader as shown in the manual of the pallet forks before attaching the log grab or assembling the hydraulic components. Follow the instructions shown on this manual to assemble the log grab.

On 200-series equipment the quick couplings and hose routing are different. Differing points about the assembly are shown separately.



Lower the pallet forks on the ground, shut down the loader engine and release any residual pressure from the hydraulic system of the loader. Follow the assembling instructions carefully.

6.1. Attaching the log grab on the pallet forks

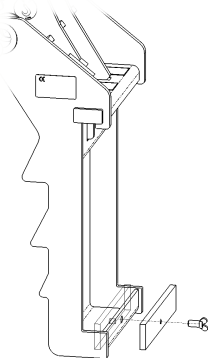


Before starting to assemble the log grab, ensure that the pallet forks are locked to the loader quick attach plate and that the lockin pin or locking pins are in the lower position and come through the fasteners of the attachment. Unlocked attachment may fall on the boom or operator and cause injuries.

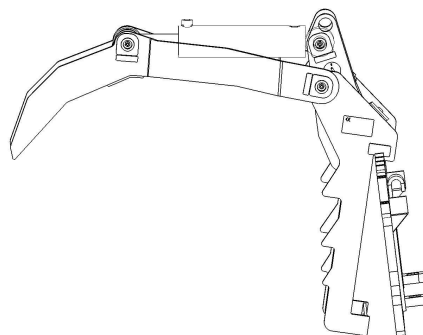
Remove the mounting bracket from the lower part of the log grab frame. If a lifting crane or other means for lifting the log grab is available, use the lifting points on top of the log grab, and lower the upper end of the log grab on the frame as shown in figures below. Align the log grab with the frame of the pallet forks. The log grab can also be lifted manually, but a safer way is to remove the forks first and then utilize the loader boom and to lift the log grab from the ground in a similar way as a regular attachment.



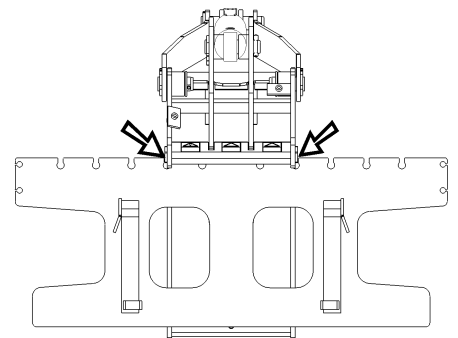
Do not lift the log grab from the grapple. As long as the hydraulic hoses and fittings are not installed and completely tightened, and until the cylinder has been de-aired, the grapple may move easily creating a crushing hazard. Beware of the moving grapple. It is advisable to have an assisting person for the assembly, as the parts are heavy.



Remove the locking bracket from the log grab



Lift the log grab to the pallet fork frame or use the loader boom to attach the log grab in a similar way as a normal attachment.

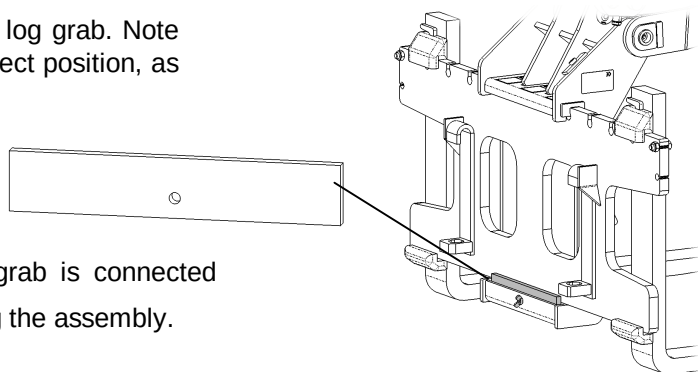


Make sure that both flanges of the log grab are aligned and the log grab is evenly on the frame plate.

Carefully install the locking bracket to the log grab. Note that the bracket should be installed in correct position, as shown in the adjacent figure.



Make sure that the log grab is connected securely before continuing the assembly.



6.1.1. Removing and storing the log grab

When the log grab is attached to the pallet forks, store the pallet forks and the log grab so that the grapple is completely lowered. Do not store the equipment directly on the ground. Use a pallet, for example.



Storage position of the log grab with pallet forks

If the log grab is stored separately, lift the grapple completely and place the log grab as shown in the figure below. Make sure that the log grab will stay stable during storage. Use load securing straps to prevent the attachment from tipping over.



Storage position of the log grab, when not attached to pallet forks

6.2. Installing the hydraulic hose kit



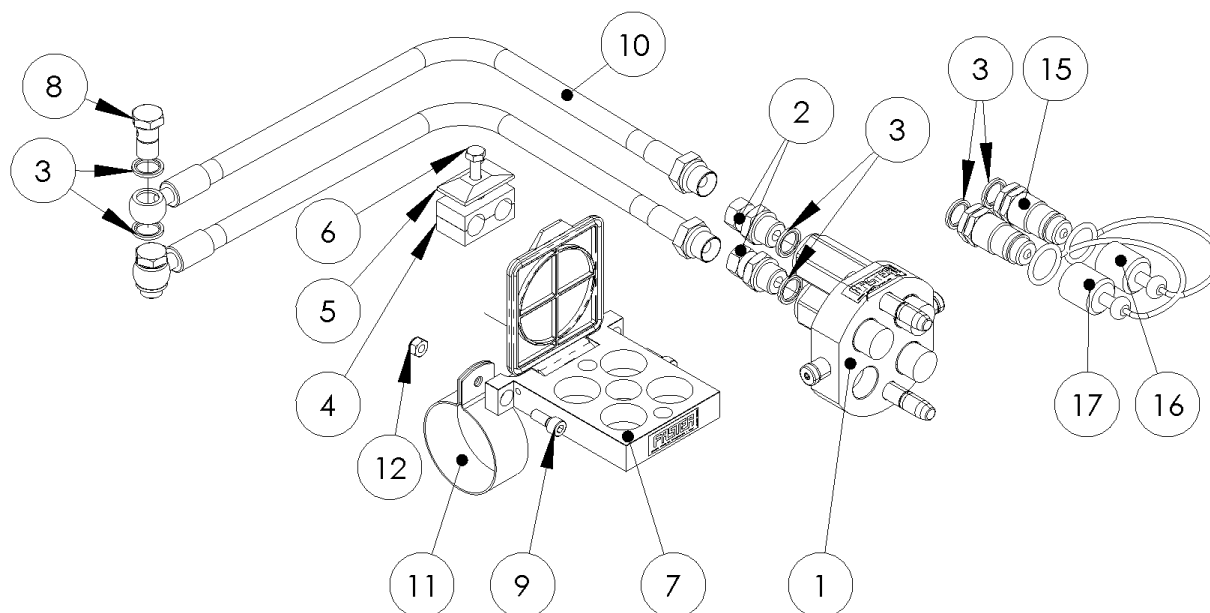
The hydraulic hoses can be assembled correctly only when the log grab has been attached to the pallet forks and they have been **attached to the loader**. Keep all hydraulic components clean during assembly.

To 300 - 700 series loaders a multiconnector system can be installed instead of the standard quick couplings. Both types of couplings are included in the hose kit A45866 until the end of 2012. Contact your dealer or nearest service point if you want to change the type of the quick couplings of your loader. On 200 series equipment only quick couplings are available.



To prevent unintentional movements of the grapple and oil spray, do not connect the quick couplings or the multiconnector to the loader before the hydraulic hoses and all fittings have been tightened.

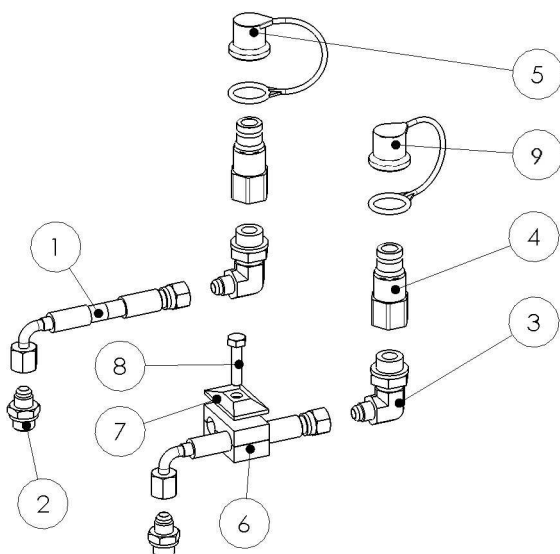
Contents of installation kit A45866 (for 300-700 series)



A45866 - Hose kit							
Ref.	Part	Qty	Description	Ref.	Part	Qty	Description
1	65251	1	Multiconnector	10	A32277	2	Hydraulic hose L=1900
2	64047	2	Connecting nipple	11	65602	2	Pipe clamp F75
3	73047	8	Usit seal ring R1/2"	12	72320	2	Nut M8 nyloc
4	64887	1	Hose clamp	13	A410342	1	Hose cover spiral
5	73767	1	Cover plate	14	A33633	1	Hose cover spiral
6	73443	1	Hexagonal screw M8*35	15	73409	2	Quick coupling nipple, 300-700 series loaders
7	65594	1	Parking for multiconnector	16	64878	1	Cover, blue, quick coupling
8	73954	2	Banjo bolt R3/8" L29	17	75753	1	Cocer, black, quick coupling
9	74481	2	Hexagonal head screw M8*20				

- The assembly kit A44701 may also be used (kit without multiconnector)
- Items 1-3 may be preassembled (or the multiconnector 65251 may include fixed fittings)
- Items 15-17 are included in the kit until the end of year 2012.

Contents of installation kit A33570 (for 200 series)



A33570 - Hose kit for 200-series			
Ref	Part	Qty	Description
1	A46557	2	Hydr. hose L=1350
2	64041	2	Basic fitting
3	64166	2	Elbow fitting
4	64563	2	Quick coupling nipple
5	64567	1	Cover, black, quick coupling
6	64886	1	Hose clamp
7	73767	1	Cover plate
8	73221	1	Hexagonal screw M8*35
9	64811	1	Cover, red, quick coupling
10	A33633	2	Hose cover spiral, 0,45 m

1. Choose a level, well lit, and clean area for installation. Lower the attachment completely on the ground or a pallet, shut down the loader engine and release any residual hydraulic pressure. Also remember safety instructions from pages 4 and 5.
2. Carefully remove the protective plastic covers from top of the hydraulic cylinder. Scrape the paint off from the contact surface. Make sure that impurities will not enter the cylinder.



When removing the plastic covers from the cylinder, the grapple will move very easily. As the grapple moves, some oil will be ejected from the cylinder. Use additional supports, e.g. blocks of wood, or lifting equipment, to prevent the grapple from moving inadvertently. Beware of the possible oil spray from the cylinder.

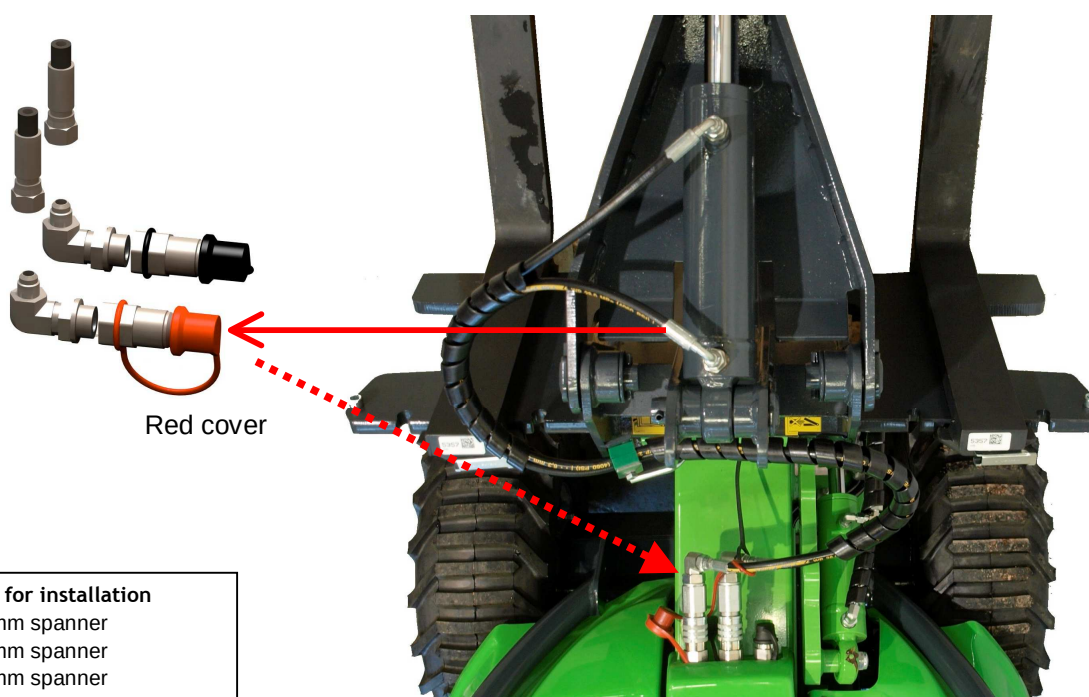
3. Install the hose kit as shown in the instructions corresponding to the correct loader type. Read instructions on page 12 about tightening the hydraulic fittings and testing the equipment.



4. A sufficient free length must be left in between the hose clamp and the hydraulic coupling of the loader, so that the log grab can be moved and tilted in all positions of the boom. Test the tightness of the hydraulic fittings and the movements of the log grab as shown on page 12.

6.2.1. Assembling the hose kit A33570, 200 series equipment

1. Install the basic fittings (part number 2) to the hydraulic cylinder and tighten well.
2. Install the elbow fittings to the quick couplings and tighten well. A seal ring should be placed in between the elbow fitting and the quick coupling.
3. Slide the protective covers to the straight ends of the hydraulic hoses and tighten carefully.
4. Fit the hoses to the hydraulic cylinder and tighten carefully to position shown in figure below. The hose fitted with a coloured protection cap should be installed to the piston side of the hydraulic cylinder, i.e. the fitting closer to the loader.
5. Connect the quick couplings to the corresponding couplings of the loader.
6. Install the hose clamp and set the hoses as shown below. If necessary, loosen the fitting between the hose and the elbow fitting, so that the hoses will settle neatly. Tighten the hose carefully. Prevent twisting of the hose with a 14 mm spanner.
7. Coil the plastic covering spirals around the hoses.
8. Test the tightness of the fittings and the operation of the log grab as shown on page 12.



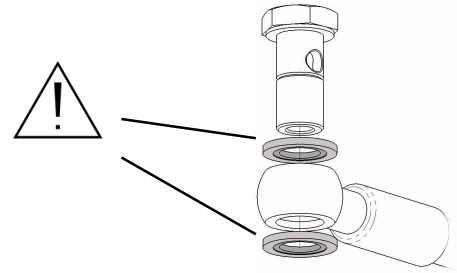
Tools needed for installation

- 14 mm spanner
- 19 mm spanner
- 22 mm spanner
- 27 mm spanner
- 32 mm spanner

200 series

6.2.2. Assembling the hose kit A45866 tai A44701, 300-700 series equipment

1. The so called banjo fittings at the end of the hydraulic hoses are connected to the cylinder with hexagonal screw type fittings. Slide a seal ring on both sides of the banjo fitting and tighten to position shown in figures. Note the welding on the cylinder and make sure that the hose is not in contact with the welding.
2. Install the hose clamp to the log grab, but do not tighten the clamp.
3. Installing the hydraulic couplings:

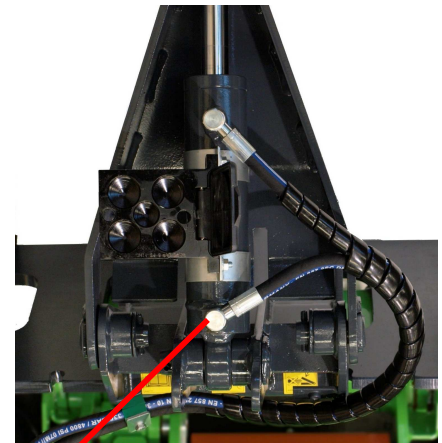


Conventional type quick couplings:

- Slide the protective caps to the straight ends of the hoses. Set a seal ring in between the quick coupling and tighten quick coupling to the hose.
- The coloured cap should be fitted to the hose connected to the piston side of the log grab, i.e. the fitting closer to the loader.

Installing the multiconnector coupler:

- Install the parking stand of the multiconnector to the hydraulic cylinder.
- Place the multiconnector fitting to the loader connector, but do not lock the connector to the loader.
- Fit the connector closer to the loader to the lower fitting of the multiconnector.

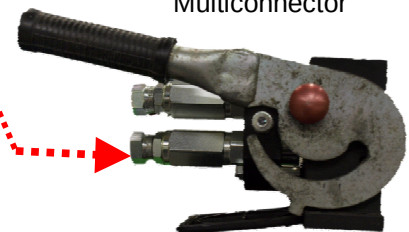


Conventional quick couplings



Blue cover

Multiconnector



Do not lock the multiconnector to the loader until the hoses are installed and all fittings have been tightened!

4. Set the hoses neatly, as shown in adjacent figure, and tighten the hose fittings well.
5. Connect the multiconnector or the conventional couplings to the loader.
6. Coil the protective spiral around the hoses and tighten the hose clamp.
7. Test the tightness of the fittings and the operation of the log grab as shown on page 12.

Tools needed for installation

- 6 mm Allen key
- 13 mm spanner
- 22 mm spanner
- 2 x 27 mm spanners
- 32 mm spanner



6.3. Tightening hydraulic fittings and testing the equipment

Tightening hydraulic fittings

Care must be taken when tightening hydraulic fittings, as overtightening of the fittings will break the fittings easily. Tighten straight fittings tightly using moderate force. Never tighten a fitting if the hydraulic system is pressurized.



Tightening the 200 series hoses and the connecting nipples of the multiconnector:

Lubricate the inner surfaces and threads of the fittings. Turn the tapered type fittings with fingers or lightly with tools as far as possible. When noticing clear increase in needed force, tighten fitting by turning 1/3 to 1/2 turns using tools. The fitting is not tightened by torque and it will not need a lot of force at any point when tightening. Overtightening will break the fitting. Try to twist the hydraulic hose by hands after tightening, the fitting should stay firm.

Test to find any leaks

- Clean the hydraulic components to find leaks more easily.
- Place a piece of cardboard under the fittings to detect leaks.
- Start the engine. While running on idle, operate the auxiliary hydraulics control lever briefly a couple of times. Shut down the engine, depressurize the hydraulics and check for any sign of leak. Tighten if necessary.
- Repeat the test by gradually increasing engine rpm and operating time. Use both directions of the auxiliary hydraulics control lever.
- At first the grapple may move inaccurately, as the cylinder and the hoses may have air trapped in them. The system will de-air itself automatically when the grapple is moved to its extreme positions.



Before checking for leakages, or going near any hydraulic fitting, to avoid possible ejection of pressurized hydraulic oil:

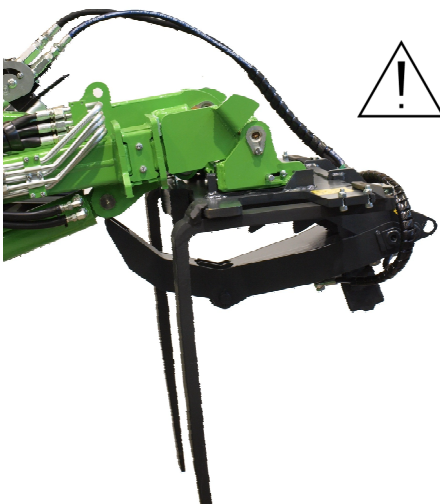
- shut down the loader engine
- depressurize the hydraulic system completely



Test the tightness of the hydraulic fittings carefully. Use idle rpm when starting the first time and keep a safe distance due to the risk of hydraulic fluid ejection. Shut down the loader and release the residual pressure before checking for leaks. Never test the tightness of the hydraulic components by placing hands near the fittings. Never tighten a pressurised hydraulic fitting, since the fitting may break and the released oil may cause serious injuries.

Test the movements of the loader boom and the attachment

After installing and testing the hydraulic components, check that the hydraulic hoses can not get stretched or pinched in any position of the loader boom or the attachment.



Carefully test the movements of the loader boom. Release the hydraulic pressure before the test. Adjust the free length of the hoses as necessary.

6.4. Connecting and disconnecting the hydraulic couplings



Never connect or disconnect quick couplings or other hydraulic components if the control lever of the auxiliary hydraulics control lever is locked on or while the system is pressurized. Connecting or disconnecting the hydraulic couplings while the system is pressurized may lead to unintended movements of the attachment or ejection of high-pressure fluid, which can cause serious injuries.

Keep the fittings as clean as possible; use the protective caps for the quick couplings on both the attachment and the loader. Dirt, ice, etc. may make using the fittings significantly more difficult. Never leave the hoses hanging on the ground; place them on top of the equipment.

Connecting standard quick couplings:

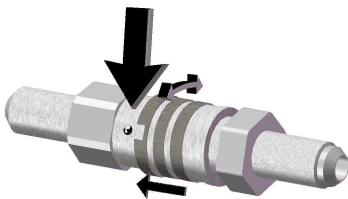
The quick couplings should be connected so that the couplings with coloured caps are connected. This way the auxiliary hydraulics control lever will move the grapple the same way every time it is used.

Before connecting or disconnecting the standard quick couplings, shut down the loader engine and move the auxiliary hydraulics lever on the loader to its extreme positions a few times. This way, there will be no residual pressure in the system and it is easier to connect the fittings.

To connect and disconnect the fittings, move the collar at the end of the female fitting. Note that the protective caps on the loader and the attachment can be fastened to each other during operation to reduce the accumulation of dirt.



In a 200-series machine, the hoses can be connected by pushing the fitting in the counterpart. When connecting and disconnecting the 200-series hydraulics quick couplings, you should pay attention to their locking. The locking prevents the female and male fittings from being disconnected from each other by accident when the attachment moves. The fitting collar may also turn by accident into the locking position, making it impossible to connect or disconnect the fittings.

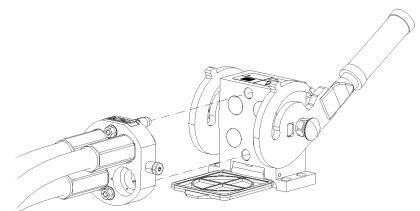


In the 200-series equipment, there is a small groove in the collar of all female fittings and a small ball below the collar. When the fittings need to be disconnected, the above-mentioned groove and ball need to be aligned. Then, the grooved collar of the female fitting is pulled downwards to disconnect the fittings from each other.

Connecting the multiconnector system:

Align the pins of the attachment connector with corresponding holes of the loader connector. The multiconnector will not connect if the attachment connector is upside down. Connect and lock the multiconnector by turning the lever towards the loader.

The lever should move easily all the way to its locking position. If the lever does not slide smoothly, check the alignment and position of the connector and clean the connectors. Also shut down the loader and release the residual hydraulic pressure.



Disconnecting the hydraulics:

Before disconnecting the fittings, lower the attachment on solid and level surface, as shown on page 8. Shut down loader engine and move the auxiliary hydraulics control lever to its extreme positions to lower the grapple and to release residual pressure.



When removing the attachment, always disconnect the hydraulic fittings before detaching the quick attach plate, to prevent hose damage and any oil spills. Reinstall the protective caps on the fittings to prevent impurities from entering the hydraulic system.

Releasing residual hydraulic pressure:

In case residual pressure is left in the hydraulic system of the attachment, it is often possible to disconnect the hydraulic couplings, but it may be difficult to connect them the next time. If the fittings will not connect, the residual pressure must be released. To release the residual pressure shut down the engine and turn the auxiliary hydraulics control lever of the loader so many times so that the grapple is freely on the ground and it will no longer move. If one of the hydraulic fittings will not connect due to the residual pressure in the system, try to lift the grapple a little at the same time as the fitting is connected.

7. Operation

Check the attachment once more before starting work.

- Check that the pallet forks and the log grab are positively attached to the frame of the pallet forks. Check that the locking bracket of the log grab is in its place and that the equipment is intact.
- Before first use, check the hydraulic fittings and the movements of the attachment as instructed on page 12.
- Check that all obstacles have been removed from the working area before operation. Remember correct working methods and avoid unnecessary descending from the loader.
- Ensure that the loader and attachment are used in a safe manner and as intended. Do not allow children to operate the machine.
- Check regularly to ensure there are no hydraulic leaks. Refer to Section 8 for maintenance instructions.
- Operate the attachment only from the driver's seat.

The log grab should be operated with the 1-pump setting of the auxiliary hydraulics.



When operating with the attachment, follow the instructions given in the operator's manual of the pallet forks and the operator's manual of the loader.



Ensure that there are no other persons near the operating area. Do not allow others closer than five meters from the log grab, to reach the boom danger area or to stay directly in front of the loader. Also make sure that it is safe to reverse with the loader. Lower the grapple and the boom before leaving the driver's seat.



When lifting material from a high level, the loader may tip forward, when starting to reverse with the loader. Do not reverse and drag the load simultaneously. Make sure that the loader can handle the load being moved.



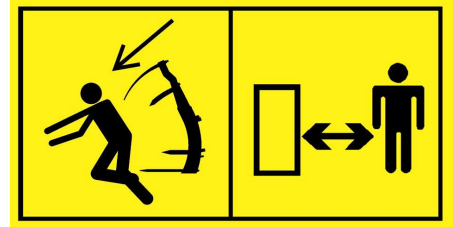
Drive carefully on slopes or inclined terrains. Always transport the load as close to the ground as possible.

8. Inspections, maintenance and servicing

Continuous maintenance of the log grab consists of regular cleaning and lubrication. The condition of the hydraulic hoses and components should be monitored regularly. Due to crushing and pinching hazards caused by the grapple, all maintenance work must be done when the grapple has been lowered down completely, the hydraulic couplings are released, and the residual pressure is released.



All maintenance procedures must be performed when the grapple has been lowered down completely. The grapple may lower by itself causing serious injuries.



Check the condition of the hydraulic hoses and components regularly when the engine has been turned off and the pressure has been relieved. Repair all leaks immediately after detecting them; a small leak can quickly grow into a big one. Do not use the equipment, if you have discovered a fault in the hydraulic system. Leaking hydraulic fluid may penetrate the skin and cause serious injuries. Hydraulic fluid is also harmful to the environment.

Check hoses visually for cracks or abrasions. Monitor the wearing of the hoses and stop the use, if the surface layer of the hose has worn off. If there are signs of leaking, to check the component, hold up a piece of cardboard in the area where a leak is suspected. Finding any of these faults means that the hydraulic hose or component in question must be replaced. Spare parts are available from your nearest AVANT retailer or service point.



Never tighten a hydraulic fitting that is under pressure, since the fitting may break and the released oil may cause serious injuries. Do not use the equipment if you have discovered a fault in the hydraulic system.



8.1. Cleaning and storing the equipment

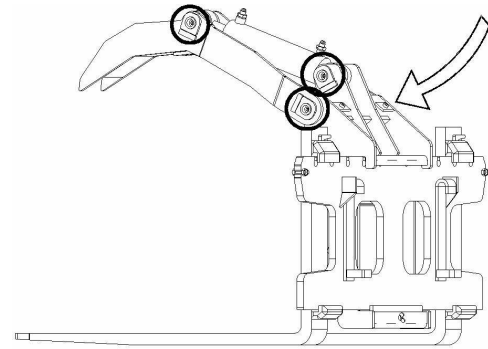
You may use a pressure washer and a mild detergent to clean the equipment. Do not use strong solvents or spray directly at the hydraulic components or at the labels on the attachment either. After washing lubricate the log grab as shown on the following page.

Ensure that the attachment is properly supported, especially when storing a log grab that is separated from the pallet forks. Do not store the attachment directly against the ground; place it on blocks of wood or on a pallet, for example. See page 8 for information about removing and storing the log grab.

For longer storage periods touch up the paint where needed, grease the lubrication points. In addition, oil the visible parts of hydraulic cylinder shaft to prevent rust damage.

8.2. Lubricating points

There are four lubrication points on the log grab, to which a small amount of lubricant is added regularly. The lubrication points are located at the both ends of the hydraulic cylinders and two more are at the grapple pivot. All lubrication nipples are standard R1/8" nipples. Replace any damaged nipples.



Lubricant must be added approximately every 1 to 3 times a year, depending on usage. Clean the end of the nipple before greasing. Only add the grease amount of a couple of grease press pushes at a time.

9. Warranty terms

Avant Tecno Oy grants a warranty of one year (12 months) for the attachment from the date of purchase.

The warranty covers repair costs as follows:

- Work costs are covered, if the repair is not performed at the factory.
- The factory replaces any defective components or consumables.

The factory may reimburse the price of components purchased by the customer in special cases that have been agreed in advance.

The warranty does not cover:

- Normal maintenance work, such as the replacement of blades or the belts, or parts and consumables required for it.
- Damages caused by unusual operating conditions or ways of use, negligence, structural changes made without the consent of Avant Tecno Oy, use of non-original parts or lack of maintenance.
- Consequences of a defect, such as interruption of work or other possible additional damages.
- Travel and/or freight costs caused by the repair.

EY-liittämisvakuutus
EG-försäkran om halvfabrikat
EC Declaration of Incorporation
EG-Einbauerklärung



Valmistaja / Tillverkare / Manufacturer / Hersteller:

AVANT TECNO OY

Osoite / Adress / Address / Adresse:

Ylötie 1

33470 YLÖJÄRVI, FINLAND

Vakuutamme täten, että alla mainitut tuotteet täyttävät konedirektiivin turvallisuus- ja terveysvaatimukset (direktiivi 2006/42/EY muutoksineen) ohjeiden mukaan asennettuna. Tuotetta ei saa ottaa käyttöön ennen kuin vaatimusten täyttyminen on todettu direktiivin mukaan. Seuraavia yhdenmukaistettuja standardeja on sovellettu:

SFS-EN ISO 12100-1, SFS-EN ISO 12100-2, SFS-EN 982

Vi försäkrar härmed att nedan beskrivna produkter överensstämmer med hälso- och säkerhetskrav i EG-maskindirektiv (EG-direktiv 2006/42/EG som ändrat) när den är monterad enligt instruktionerna. Produkten får inte tas i bruk förrän överensstämmelse med direktivens bestämmelser har konstaterats. Följande harmoniserade standarder har tillämpats:

SFS-EN ISO 12100-1, SFS-EN ISO 12100-2, SFS-EN 982

We hereby declare that the products listed below are in conformity with the provisions of the Machinery Directive (directive 2006/42/EC as amended) when installed according to the instructions. The product must not be taken in use before conformity with the directive has been stated. The following harmonized standards have been applied:

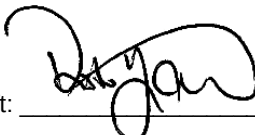
SFS-EN ISO 12100-1, SFS-EN ISO 12100-2, SFS-EN 982

Wir erklären hiermit, dass die nachstehend aufgeführte Maschine konform ist mit den Bestimmungen der EG-Maschinenrichtlinie (EG-Richtlinie 2006/42/EG mit Änderungen) wenn sie gemäß den Anweisungen eingebaut worden ist. Das Produkt darf nicht in Betrieb genommen werden, bevor die Konformität mit der Richtlinie festgestellt worden ist.

Die folgende harmonisierte Normen wurden angewandt:

SFS-EN ISO 12100-1, SFS-EN ISO 12100-2, SFS-EN 982

Kuvaus: Avant-trukkihaarukoiden kanssa käytettävä puukahmari A34822/A33503
 Beskrivning: Timmergrip A32822/A33503 för användning med Avant-palgaffeln
 Description: Log grab for A34822/A33503 use with Avant pallet forks
 Beschreibung: Holzgreifer A34822/A33503 für die Verwendung mit Avant Palettengabel

Allekirjoitus / Namnteckning / Signature / Unterschrift:  10.6.2010 Ylöjärvi, Finland

Risto Käkelä,
 Toimitusjohtaja / Verkställande direktör /
 Managing Director / Geschäftsführer





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